

Mario Alejandro Tabares Arango

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PROFESSIONAL SUMMARY

AI/ML engineer specializing in AI systems that must withstand regulatory, audit, and public scrutiny. Built and defended 230+ production risk models across four countries (USD 14B exposure) through eight audit cycles with zero material findings. Built graph RAG and knowledge-graph systems in government serving 2.5M residents, including PII pipelines protecting 65,000+ people. Seven years in production ML and model risk, the last two on generative AI. MS in AI Systems Management, Carnegie Mellon.

EDUCATION

Carnegie Mellon University, Heinz College

Pittsburgh, PA

Master of Science in Artificial Intelligence Systems Management.

Expected December 2027

Relevant Coursework: Machine Learning, Causal Inference & Program Evaluation, Applied Econometrics, AI Governance & Risk Management, Data-Driven Decision Making

University of Antioquia

Medellin, Colombia

Bachelor of Science, Philosophy. Thesis: "How to Say No: Modeling the Precautionary Principle" (Honorable Mention, Nov. 2023).

December 2024

National University of Colombia

Medellin, Colombia

Bachelor of Science, Physics Engineering. Thesis: "Computational Spaces".

September 2019

EXPERIENCE

Associate Director of Data and Technology

January 2025 – July 2026

CORLIDE (Corporación Libertad y Democracia) - Colombian nonprofit for institutional transparency

Medellin, Colombia

- Built a graph RAG system using Neo4j, Pinecone, and LightRAG over three years of 500+ anonymized citizen petitions, driving overdue responses (>15 business days) from 300+ to zero per month
- Designed a domain ontology released as a public asset¹, standardizing entities and relationships as a semantic layer for downstream AI systems, reducing hallucinations about conceptual misunderstandings from 55% to 25%
- Cut complex spatial-question turnaround from 1 analyst-week to 10 minutes (240×) by deploying geospatial AI agents, validated against a purpose-built 96-question execution-accuracy benchmark spanning 34 historical data standards
- Created and implemented a PII anonymization pipeline protecting 65,000+ individuals and 400,000+ addresses, stripping 13 sensitive fields via irreversible hashing (SHA-256), suppression, and generalization
- Led AI and data governance for the Undersecretariat of Public Space, Mayor's Office of Medellin (2.5M+ residents), implementing DAMA-DMBOK and ITIL frameworks to clean and integrate 250+ data sources
- Coordinated with the Open Government Partnership, Colombia's Presidency, and Colombian universities (Externado, Los Andes) on a national AI governance model; partnered with the UK's FCDO on Medellin's Local Open Government strategy

Senior Data Scientist

July 2020 – January 2025

Bancolombia (Grupo Cíber) — Colombia's largest bank, NYSE: CIB

Medellin, Colombia

- Served as primary technical counterpart to PwC México (external audit), Internal Audit, Statutory Audit, and Colombian banking regulators across 8 audit cycles covering 230+ credit risk models — closed every cycle with zero material findings, in an NYSE-listed, SEC-reporting environment (Form 20-F)
- Designed and built 230+ loan-loss provisioning models (lifetime expected losses) under IFRS 9 — the international counterpart to US CECL — using Aalen-Johansen multistate survival models with time-dependent covariates, Cox proportional hazards models, and semi-Markov multistate chains; led a team of 3 on evaluation and data-cleaning methodology
- Owned the provisioning model suite across four countries (Colombia, Panama, El Salvador, Guatemala), covering ~COP 56T (USD 14B) in credit exposure and 14M+ borrower records

¹ <https://bottomuplabs.github.io/ep-ontology/>

- Applied Monte Carlo and Bernoulli-based simulation to assess loss behavior and concentration risk, expanding backtesting (S2+) from 20% to 90% and strengthening early risk identification
- Re-architected the model-development data layer as a change-based feature store: pulled from Impala over the enterprise warehouse and persisted only state transitions into InfluxDB, compressing ~1 TB of borrower history into 16 GB and eliminating a 3-hour dataset rebuild per model
- Built a lightweight experiment registry — per-run SQLite state files versioned with DVC, metrics tracked in MLflow — enabling parallel execution at 2,000 runs/week vs. 20/month and systematic search over model–variable combinations; resulting specifications cut PD prediction error (relative RMSE of predicted PD vs. realized default rates) from 30% to 5% (natural-person) and ~50% to under 10% (large-corporate)

Data Analyst

March 2019 – July 2020

Telemetrik

Medellin, Colombia

- Led a data-architecture transformation enabling real-time analytics for 2000+ IoT devices (water sensors, streetlights, sweep-tracking) and automated dashboards, reducing manual monitoring of service interruptions supporting Smart City analytics
- Implemented automated data-quality controls, including range checks, null-rate monitoring, and statistical outlier detection to identify sensor failures and data corruption across 10+ data streams

SKILLS

AI Systems: Retrieval-Augmented Generation (RAG), Knowledge Graphs, Neo4j, Pinecone, LightRAG, Ontology Design, Machine Learning, SHAP, Explainable AI

Programming & Data: Python, SQL, Apache Spark, Impala, Hive, Hadoop, dbt, Git

Model & AI Risk: Model Risk Management, Model Validation, Model Monitoring, AI Governance, AI Risk Management, Statistical Modeling, IFRS 9, SOX

Languages: Spanish (Native), English (Full Professional, IELTS C1), French (Full Professional, DELF B2)

CERTIFICATIONS

- Data Science for All (DS4All) - CorrelationOne, 375+ hours, completed December 2019

AWARDS & HONORS

- "Being EXTRAordinary" Team Recognition for leading performance - Grupo Cibest (CIB) (2024)
- Honorable Mention, Philosophy Thesis - University of Antioquia (2023)
- Emerging Leaders in the Americas Program (ELAP) - Government of Canada Scholarship (2018)

TEACHING EXPERIENCE

- University of Antioquia, Faculty of Economics, Medellin, Colombia | Faculty | January 2025 - June 2025 Taught a 20-student undergraduate Data Science course; developed exams and assignments and reviewed curriculum for accreditation standards
- National University of Colombia, Faculty of Humanities, Medellin, Colombia | Faculty | July 2024 - December 2024 Taught Apache Spark, the Hadoop ecosystem, and Impala within an undergraduate Big Data Engineering course for 60 students

RESEARCH & PUBLICATIONS

- "Zombie Governance" - Extended abstract accepted for presentation, Data for Policy 2026 (international conference on data science and public policy)
- "How to Say No: Participatory Deliberation in the Precautionary Principle" - Presented, International Congress of Philosophy: Justice, Recognition and Peace, August 2023
- Researcher, Knowledge, Philosophy, Science, History and Society research group, University of Antioquia (August 2020 Present)